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(54) **METHOD AND SYSTEM FOR IN-LINE
OPTICAL SCATTEROMETRY**

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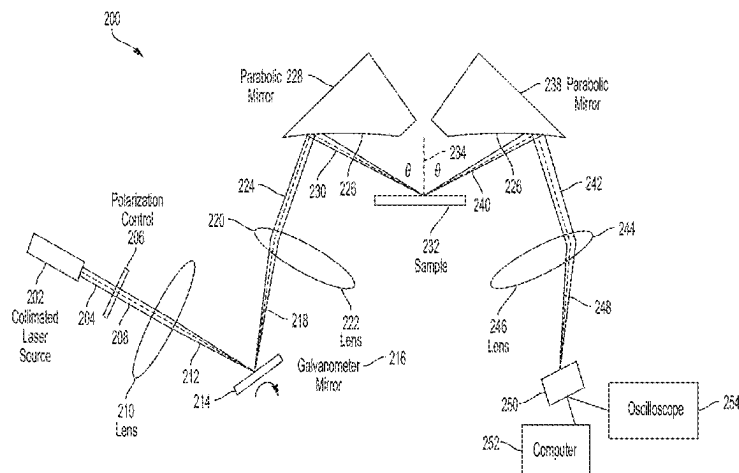
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ABSTRACT

A system for measuring a periodic array of structures on a sample is provided. The system includes an optical source configured to produce an optical beam; an optical system configured to control the polarization of the optical beam and to focus the optical beam with a first NA_1 on a sample surface and to sweep the angle of incidence across a range of angles with an approximately fixed focal position on a sample surface with a second NA_2 wherein $NA_2 > NA_1$; additional optical components configured to receive the optical beam reflected from the sample surface and to focus the reflected beam onto a detector; and a recording system to record the reflectivity of the sample surface as a function of the angle of incidence. In an embodiment, the optical system provides a spot on the sample such that both the

(Continued)



angle of incidence and the position on the sample are varied during a sweep. Electronic filtering is provided to separate low frequency signals, corresponding to structural details of the sample, and high frequency signatures, corresponding to localized defects on the sample.

15 Claims, 9 Drawing Sheets

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(51) Int. Cl.

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See application file for complete search history.

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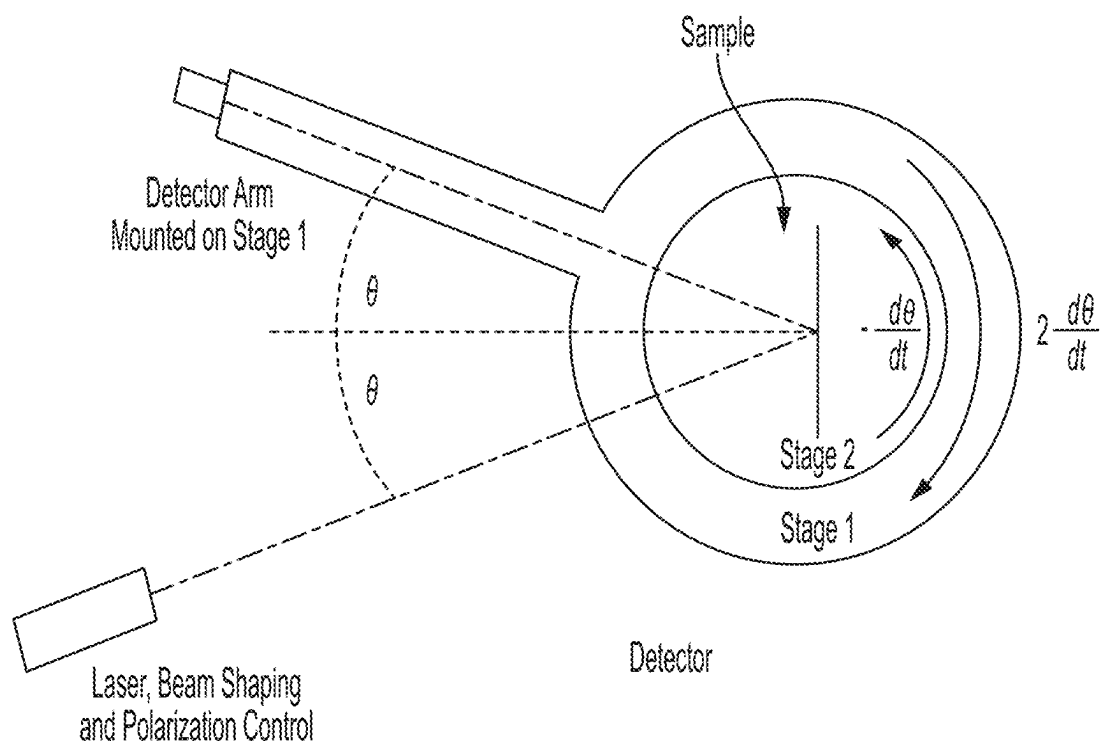
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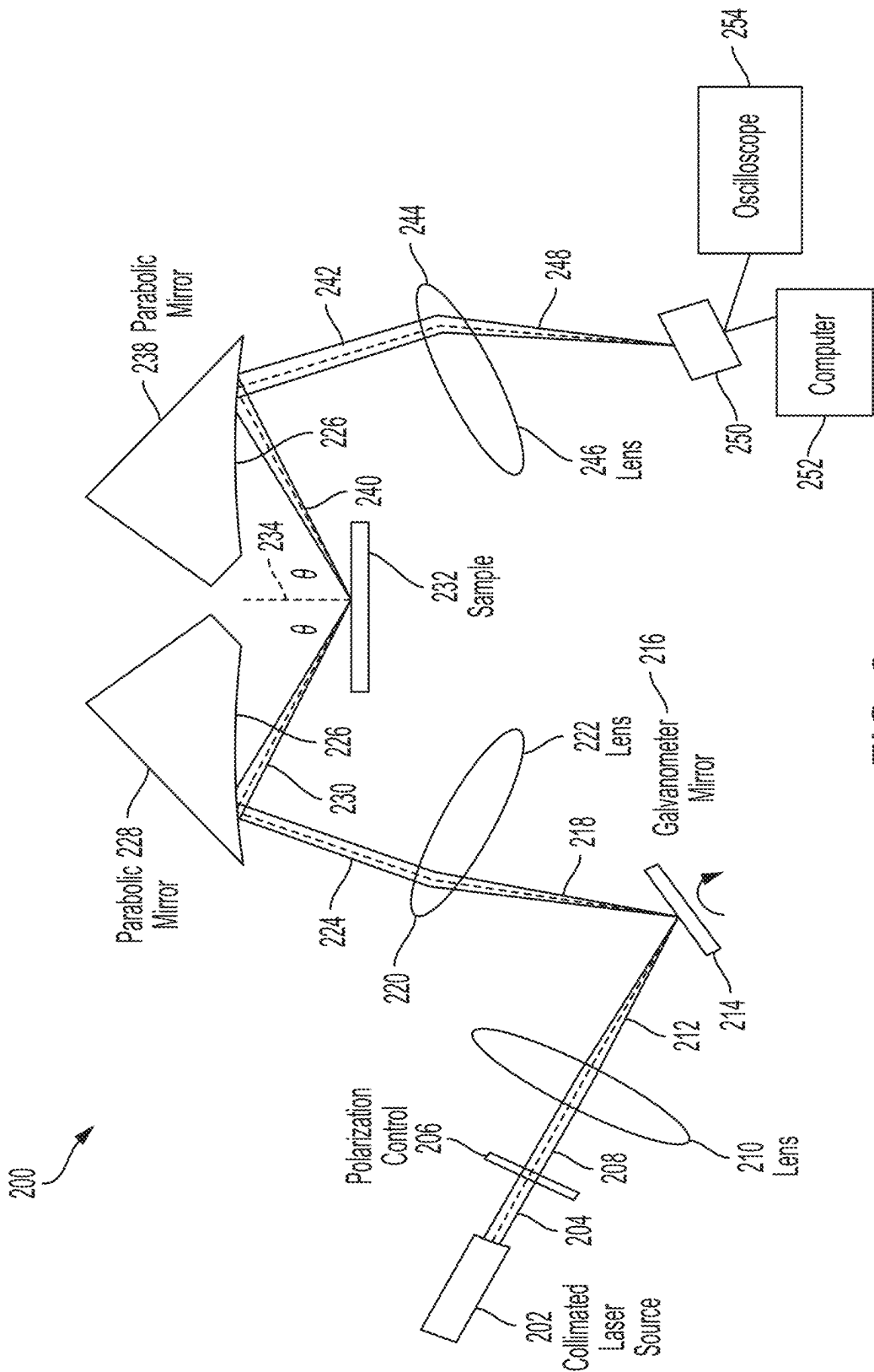


FIG. 2

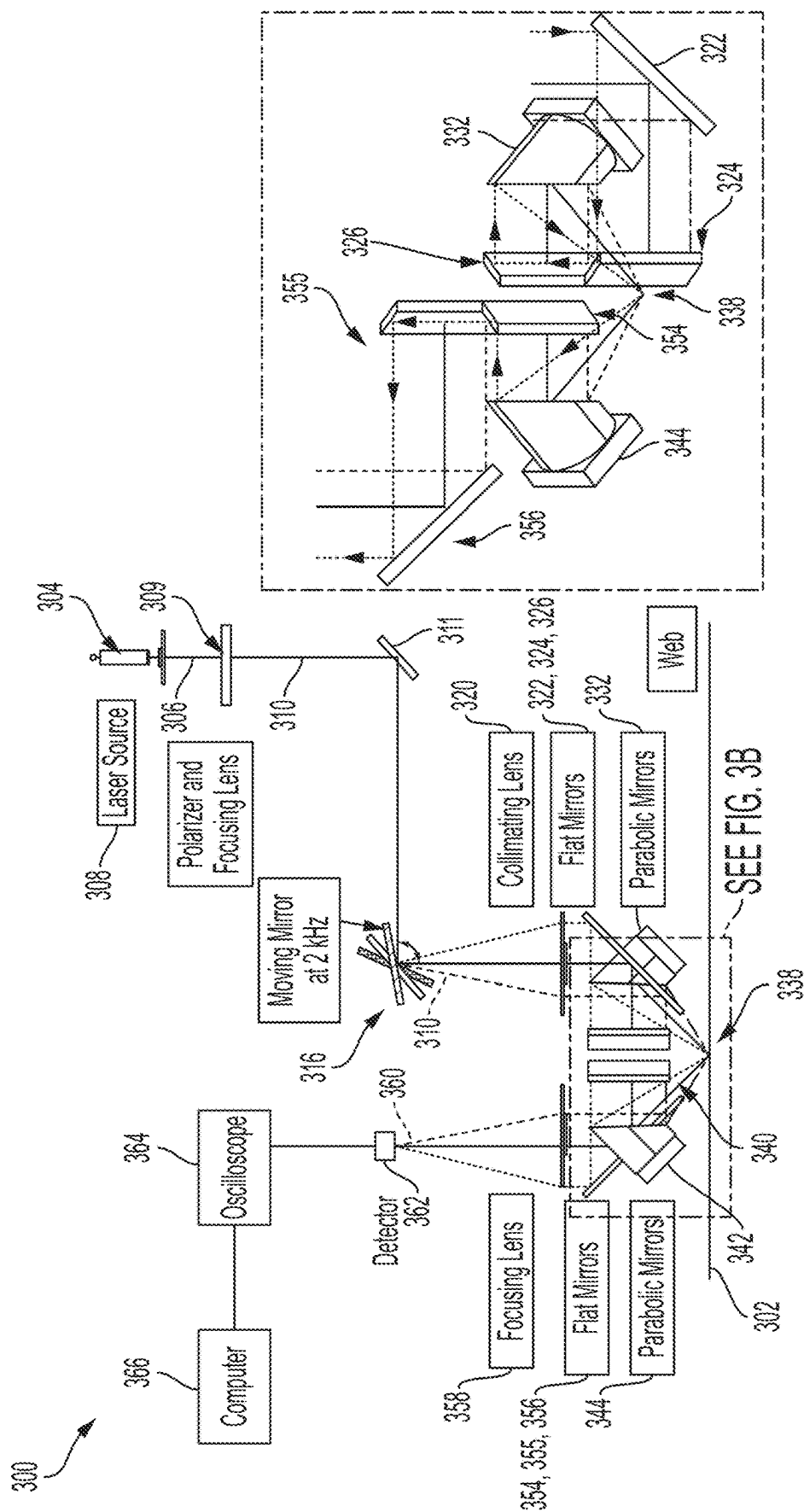
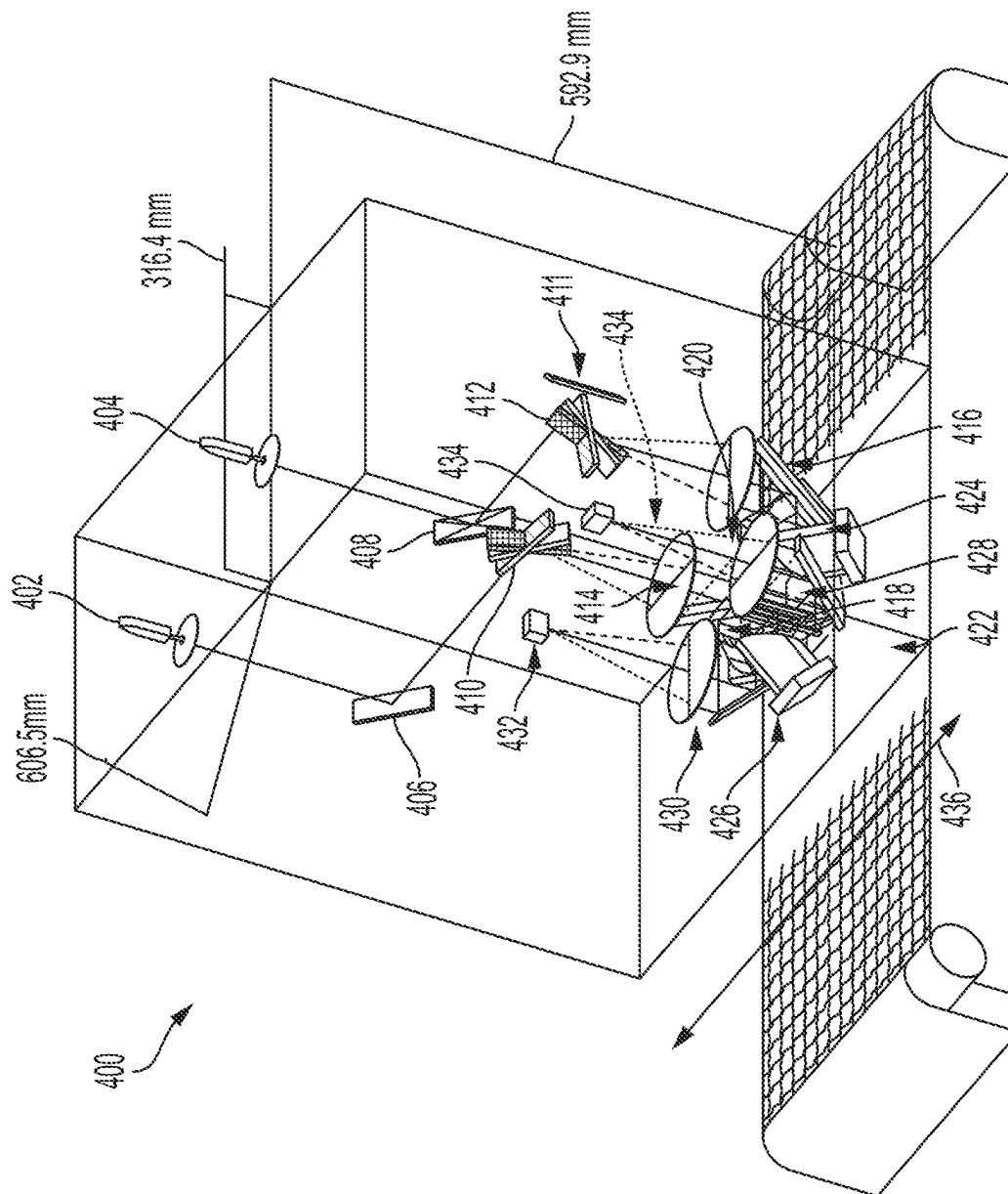


FIG. 3A

FIG. 3B



45E

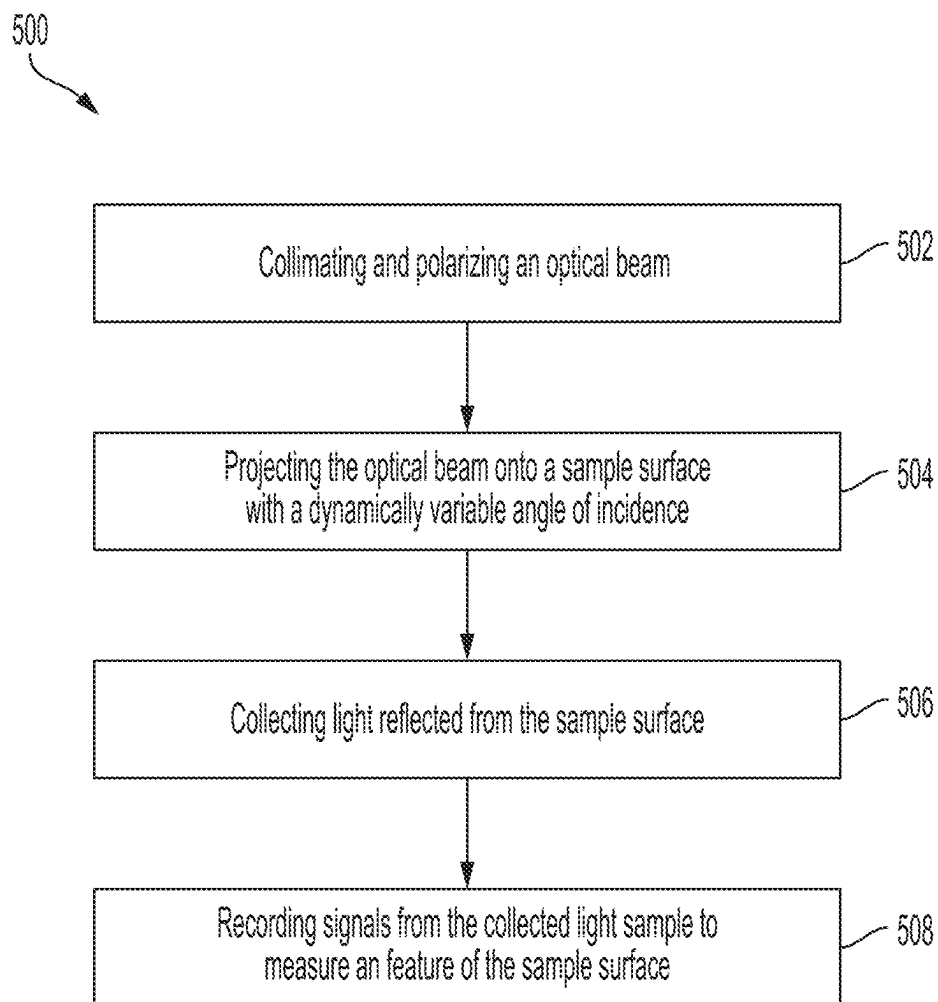


FIG. 5

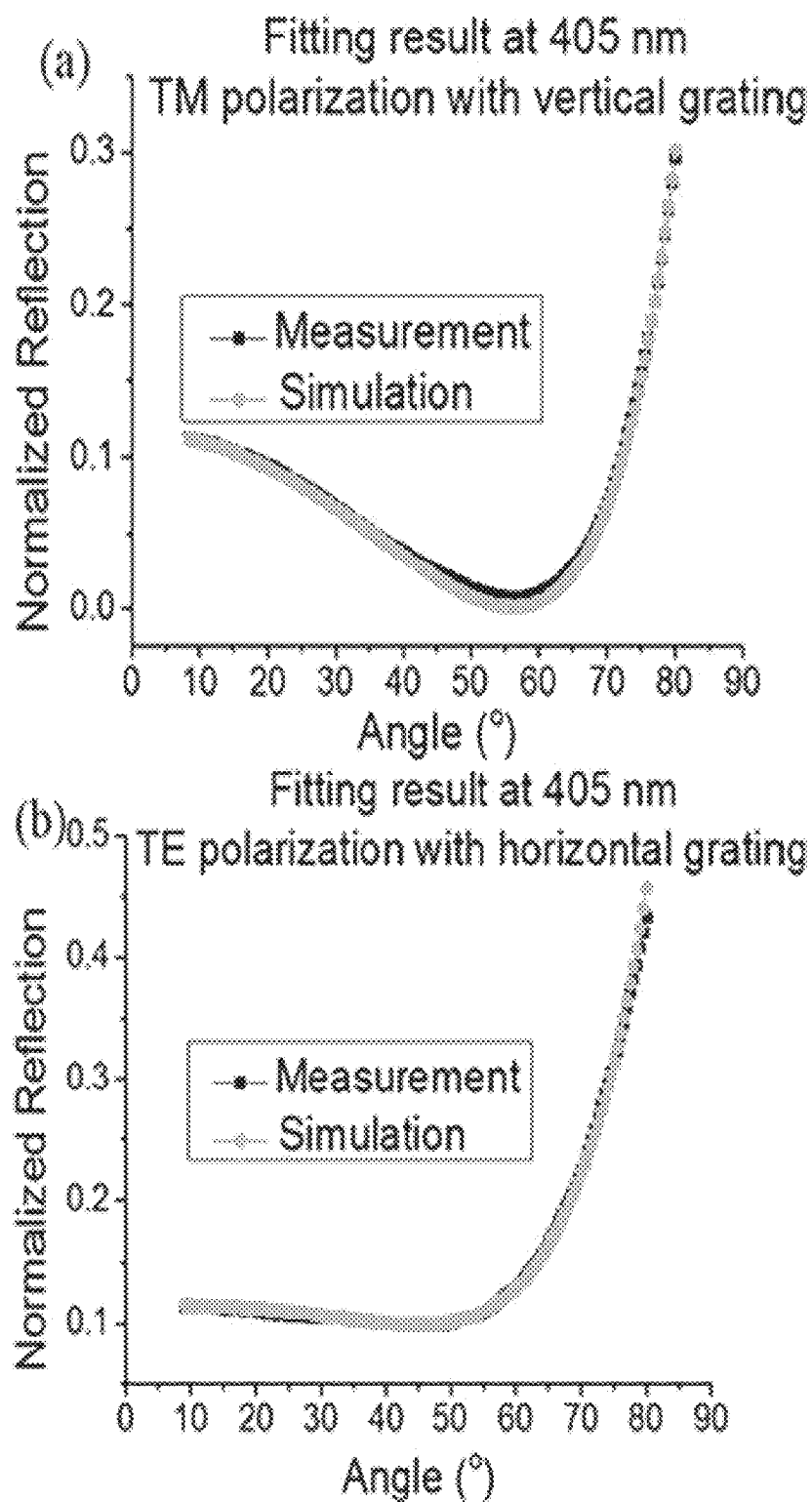


FIG. 6

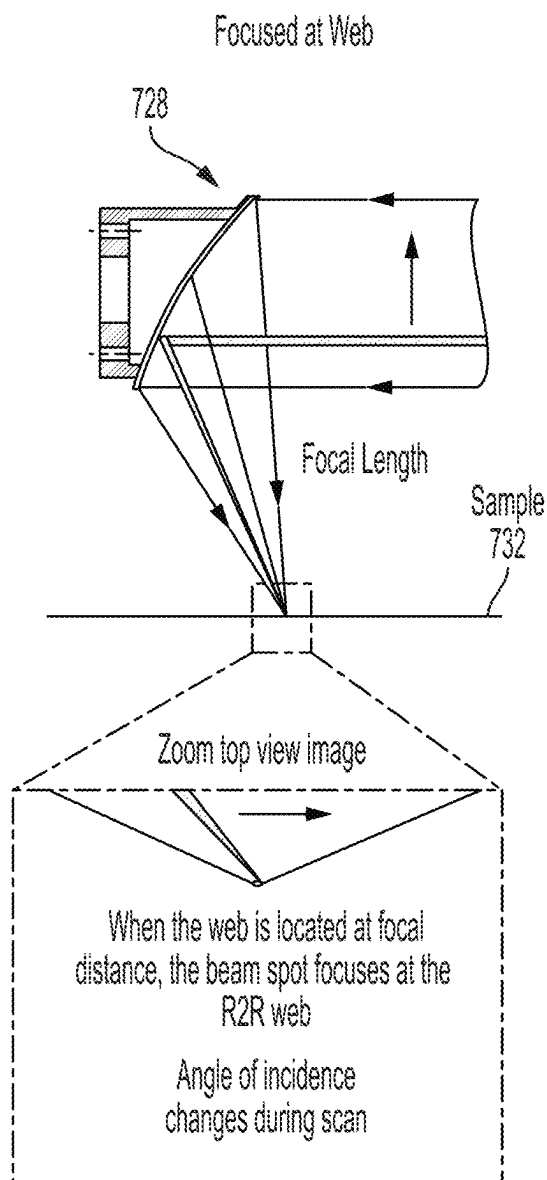


FIG. 7A

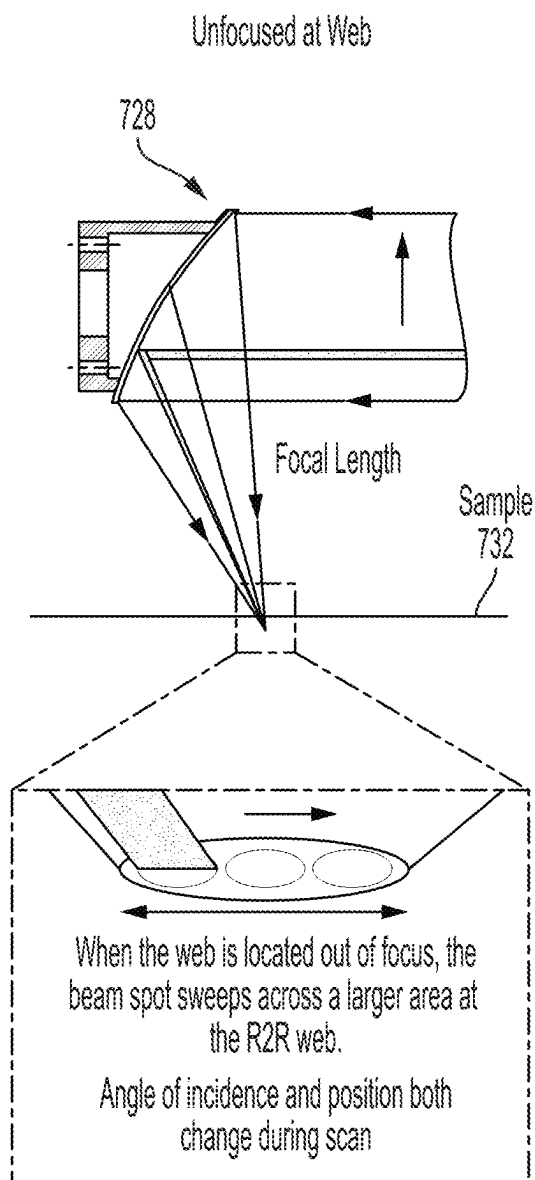


FIG. 7B

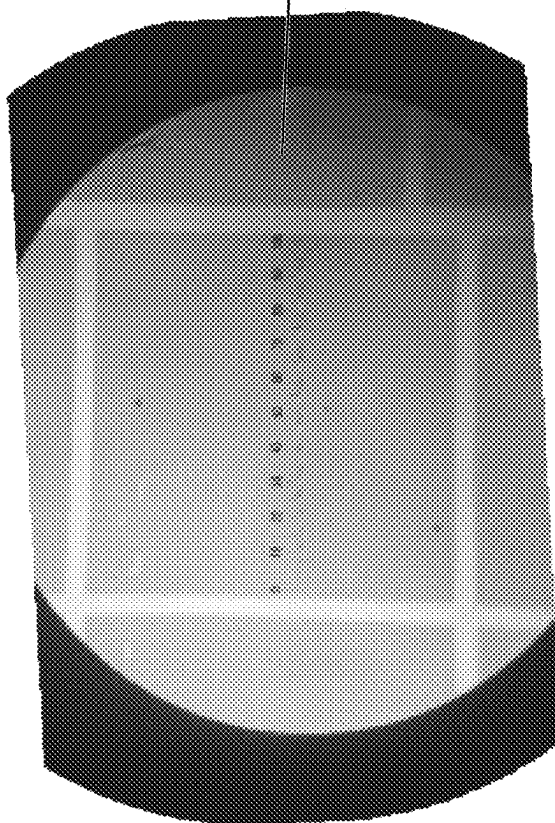


FIG. 8A

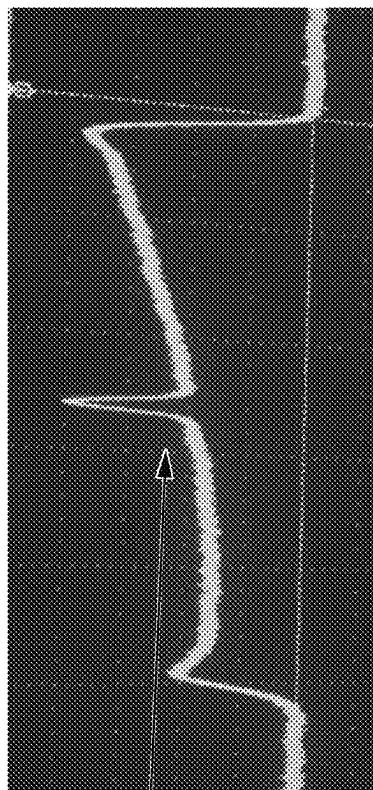


FIG. 8B

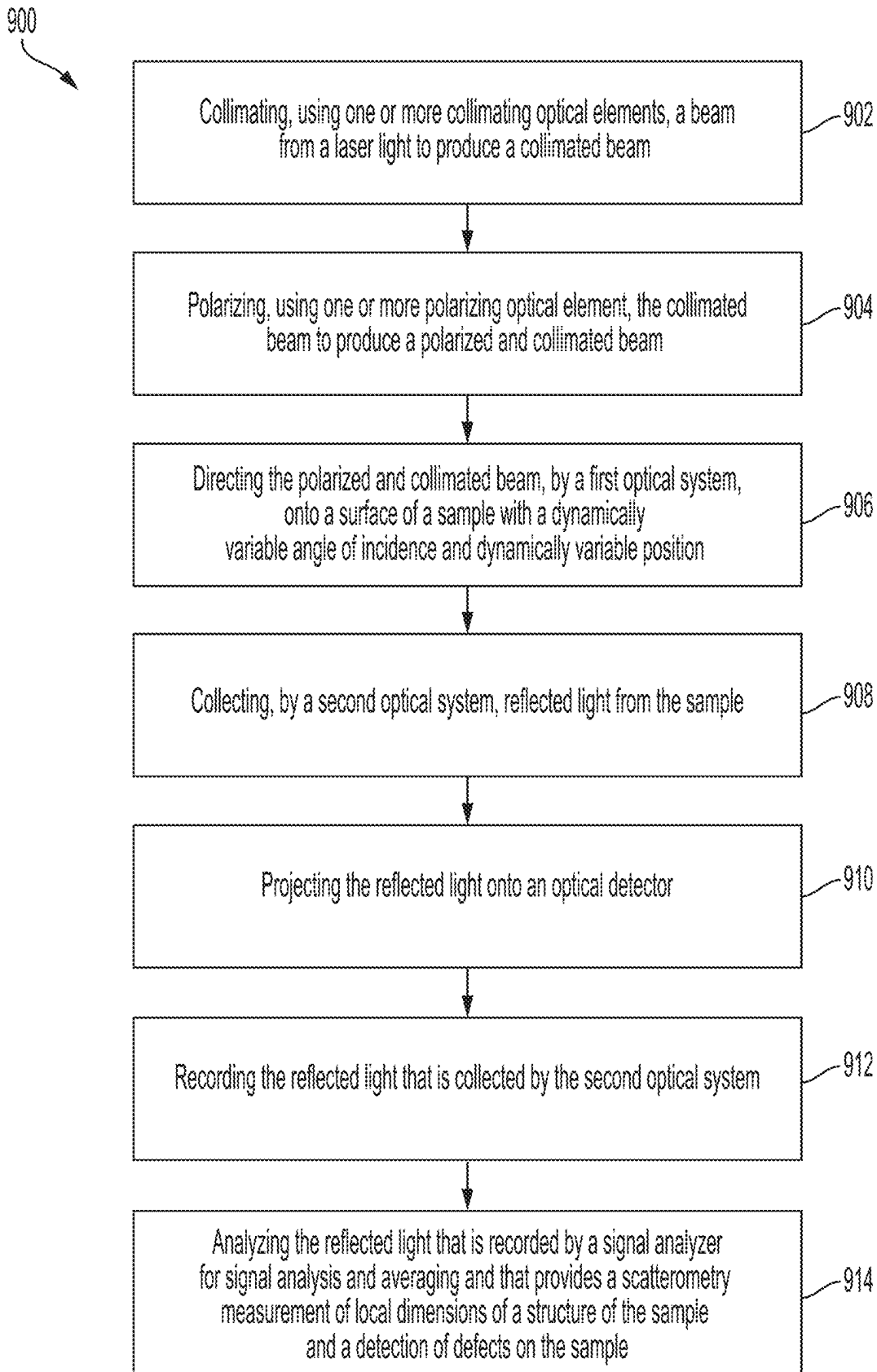


FIG. 9

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METHOD AND SYSTEM FOR IN-LINE OPTICAL SCATTEROMETRY

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation-in-part of U.S. patent application Ser. No. 17/053,585 filed on Nov. 6, 2020, which is a 371 National Phase application based on International Application PCT/US2019/030943 filed May 6, 2019, and it claims priority to U.S. provisional patent application No. 62/668,003 filed on May 7, 2018 and U.S. provisional patent application No. 62/776,640 filed on Dec. 7, 2018, which are both hereby incorporated by reference in their entireties.

GOVERNMENT FUNDING

This invention was made with government support provided by the National Science Foundation under grant No. 2041676. The Government has certain rights in the invention.

BACKGROUND

Metrology is essential to any manufacturing process. The challenges for metrology are increasing as the scale of manufacturing is decreased to micrometer and even nanometer linear dimensions in many product areas such as electronics and photonics. Metrology can be off-line (in a separate tool), at-line (in a tool associated with the manufacturing process), or in-line (in a tool that monitors the manufacturing process at speeds comparable to the manufacturing operation and without disruption of the manufacturing flow). The requirements for speed of the metrology increase as the metrology unit process is brought closer to the manufacturing line. For an in-line tool, the measurement speed has to be commensurate with the speed of the processing tool, for example, web speed for a roll-to-roll machine. As well as speed, the requirements for an in-line metrology include: resolution commensurate with the manufacturing scale; areal coverage sufficient to provide information relevant to manufacturing yield; nondestructive operation, and ideally noncontact, so as not to impact the yield.

Current semiconductor devices are fabricated at a linear scale of 10 nm or below and employ structures such as 3D ridges (FINFET architecture) with complex 3D topologies. Due to the increased complexity of semiconductor devices and processing, increasing numbers of metrology steps are required for monitoring and quality control; which translate to expensive and time-consuming processes during fabrication.

Two techniques that are often used for dimensional metrology in the semiconductor industry are scanning electron microscopes (SEM) and atomic-force microscopes (AFM). SEM requires a high vacuum environment which limits its applicability for in-line applications and top-down SEM measurements cannot provide information on the 3D structure. Cross-section SEM can provide this information; however destructive sectioning of the substrate is required. Additionally, each sample only examines a few structures and it is difficult to sample multiple regions of a large wafer. AFM can be applied to a full wafer, but usually only covering small sampled areas, however, there are significant difficulties involved with deconvolution of the tip topology

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from the measured lineshape as well as issues related to the robustness of the tip and the speed of measurement.

Scatterometry, the measurement of light reflected/transmitted/scattered from a sample has a long history of providing metrological information. Scatterometry is a noncontact, nondestructive far-field optical technique that satisfies most of the metrology requirements. Sensitivity to nanoscale feature dimensions has been demonstrated. Two directions in scatterometry have been investigated: 1) ellipsometric scatterometry where the complex polarization response (Mueller matrix) is measured as a function of wavelength of the incident light using a broadband incoherent source; and 2) angular scatterometry where the angle of incidence is varied at a fixed wavelength using a laser source. Each has advantages: ellipsometric scatterometry has greater sensitivity, but a) requires a more detailed knowledge and control of the properties of all of the involved materials as a function of wavelength and b) requires a long measurement time because of restrictions on optical power density imposed by an extended, incoherent source. Angular scatterometry has a potential for a faster response since it uses a much brighter [power/(wavelength-cm²-solid-angle)] coherent laser source that can be focused to small dimensions as needed by metrology targets and offers much higher power densities, allowing larger bandwidth receivers to accommodate higher speeds. Additionally, angular scatterometry only requires knowledge of the optical properties at a single wavelength, usually well removed from spectral regions of high material variability, which makes the interpretation of the scattering signature vs. angle more reliable.

2θ scatterometry, a noncontact optical measurement of the reflectivity of a structured pattern on a wafer, has demonstrated the necessary resolution for 3D structures. However most designs operate as off-line tools with measurement times that are incompatible with manufacturing process control needs.

The silicon industry is dominated by wafer-scale processing where stand-alone metrology tools are commonly used, and metrology measurements are not necessarily correlated with the manufacturing cycle. Increasingly, there is intense interest in migrating micrometer and sub-micrometer scale manufacturing to roll-to-roll (R2R) processes that offer significant throughput advantages. For R2R manufacturing, it is necessary to include metrology that can operate at speed commensurate with the manufacturing process. This requires a measurement speed such that the R2R web is effectively stationary during the measurement.

Additionally, many implementations of 2θ scatterometry involve synchronous mechanical rotation of both the sample and a detector in order to track the reflectivity as a function of angle. This is incompatible with R2R processes where the sample (web) is continuously moving in one direction and cannot be manipulated for the measurement.

Thus, a heretofore unaddressed need exists in the industry to address the aforementioned deficiencies and inadequacies.

SUMMARY

According to examples of the present disclosure, a system for measuring a periodic array of structures on a sample is provided. The system comprises an optical source configured to produce an optical beam; an optical system configured to control the polarization of the optical beam and to focus the optical beam with a first NA₁ on a sample surface and to sweep the angle of incidence across a range of angles with an approximately fixed focal position on a sample

surface with a second NA_2 wherein $NA_2 > NA_1$; additional optical components configured to receive the optical beam reflected from the sample surface and to focus the reflected beam onto a detector; and a recording system to record the reflectivity of the sample surface as a function of the angle of incidence.

In some examples, the optical source comprises a laser. In some examples, the optical system comprise a movable mirror configured to sweep the optical beam across a range of angles of incidence at a frequency of about 2 kHz; a lens configured to collimate an expanding optical beam reflected from the movable mirror, and optical components configured to focus the optical beam that is collimated onto the sample surface at a variable angle of incidence. In some examples, the optical components comprise a parabolic mirror. In some examples, at least one 2 θ scatterometry apparatus is integrated into a roll-to-roll manufacturing tool. In some examples, the output of the 2 θ scatterometry measurement is used to provide a metrology module for assessing the performance of the manufacturing tool.

According to examples of the present disclosure, a method for 2 θ scatterometry is provided. The method can comprise collimating and polarizing, using optical components, an optical beam from a laser light source to produce a collimated and polarized optical beam; projecting, using additional optical components, a focal spot of the collimated and polarized optical beam onto a first surface of a sample with a dynamically variable angle of incidence; collecting, by a third optical system, light reflected from the sample at an optical detector; recording a signal from the optical detector giving the reflectivity vs. angle of incidence of the sample; and interpreting the recorded angular spectrum to monitor the status of the sample.

In some examples, the second optical system projects a spot size of about $240 \times 220 \mu m^2$ at the sample surface. In some examples, sweeping the laser light source across a surface of a parabolic mirror provides an angular range of about 15° to about 65° . In some examples, the 2 θ scatterometer is incorporated into a roll-to-roll manufacturing tool. In some examples, the method can further comprise recording a plurality of signals from the optical detector and averaging the plurality of signals.

According to examples of the present disclosure, a system for 2 θ scatterometry is provided. The system can comprise a laser light source configured to provide a laser beam; one or more control optics configured to collimate and polarize of the laser beam; a first optical system configured to project a focal spot of the laser beam onto a first surface of a sample with a dynamically variable angle of incidence; a second optical system configured to collect reflected light from the sample and project collected reflected light onto an optical detector; and a digital oscilloscope configured to average and record a signal from the optical detector.

In some examples, the system can further comprise a galvanometer; and a mirror mounted on the galvanometer configured to sweep the focal spot of the laser light across the first surface of the sample. In some examples, the first optical system reduces a spot size of the laser beam to about $240 \times 220 \mu m^2$ at the sample surface. In some examples, the galvanometer provides a sinusoidal oscillation of about 2 kHz or more and a scan duration of about 0.25 ms/scan or less.

In some examples, the sample contains an patterned array of structures. For a pattern with symmetry in one-dimension pattern (e.g. a grating), the angular sweep can be in a direction parallel or perpendicular to the grating lines. For some cases, additional information is available for other

directions of rotation, for example at 45° to the grating lines. For a pattern with symmetry in two-dimensions, the angular sweep can be in directions parallel to one or the other of the symmetry axes. In some cases, additional information is available for other directions of rotation.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the present teachings and together with the description, serve to explain the principles of the disclosure.

FIG. 1 shows a conventional 2 θ -scatterometer apparatus including two concentric rotation stages.

FIG. 2 shows a first 2D design in-line optical scatterometry set up according to examples of the present disclosure.

FIG. 3A shows a second 2D design in-line optical scatterometry set up according to examples of the present disclosure.

FIG. 3B shows a perspective view of a portion of the optical set-up of FIG. 3A showing the mirror arrangement near the parabolic mirrors.

FIG. 4 shows a 3D design for an in-line optical scatterometry set up based on the second design of FIGS. 3A and 3B where one scatterometer is arranged along the web direction and one scatterometer is arranged across the web direction.

FIG. 5 shows a method for measuring a feature of a sample using a 2D design in-line optical scatterometry of FIG. 2 or FIGS. 3A and 3B according to examples of the present disclosure.

FIG. 6 shows a fitting result for a wire-grid polarizer using a 405 nm laser source. The fitted parameters of the Al grating lines are: pitch of 97.2 nm, bottom linewidth (at substrate) of 49.4 nm; top width (before rounding) of 58.6 nm; aluminum height of 208.3 nm; undercut into fused silica of 22.9 nm; horizontal rounding of 25.8 nm; vertical rounding of 14.8 nm and Al_2O_3 overcoat of 4 nm. The apparatus of FIG. 1 was used to obtain this measurement.

FIG. 7A and FIG. 7B show sweeping across the R2R web by changing focal length from parabolic mirrors according to examples of the present disclosure.

FIG. 8A and FIG. 8B show observation of a defect during an in-line optical scatterometry measurement as the optical beam is swept across a $50 \mu m$ wide non-filling defect according to examples of the present disclosure.

FIG. 9 shows a method according to examples of the present disclosure.

DETAILED DESCRIPTION

Reference will now be made in detail to the present embodiments, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

FIG. 1 shows a conventional 2 θ -scatterometer apparatus including two concentric rotation stages. This tool requires: 1) mounting a small piece of the sample with the measurement surface on the common rotation axis of the two stages; and 2) coordinated rotation of the stages to vary the angle of incidence while maintaining the reflected beam on the detector. Measurement time including signal averaging is several minutes, which is suitable for off-line measurements but is clearly too long for an in-line roll-to-roll tool that has to operate on time scales commensurate with the web speed. The scatterometry set-up demonstrated in FIG. 1 involves

two computer-controlled rotation stages. See for example, R. Zhu, S. R. J. Brueck, N. Dawson, T. Busani, P. Joseph, S. Singhal and S. V. Sreenivasan, *Scatterometry for Nanoprint Lithography*, J. Vac. Sci. Technol. B34, 06K503 (2016).

The silicon industry is dominated by wafer-scale processing where stand-alone metrology tools are commonly used, and metrology measurements are not necessarily correlated with the manufacturing cycle. Increasingly, there is interest in migrating micron and sub-micron scale manufacturing to roll-to-roll (R2R) processes that offer throughput advantages. For R2R manufacturing, the inclusion of metrology that can operate at speed commensurate with the manufacturing process would be beneficial. Accordingly, aspects of the present disclosure provide for a 2θ-scatterometer apparatus and a method of operating the 2θ-scatterometer apparatus to monitor structures on a sample is provided that allows an optical beam to be moved while the sample stage is stationary (on the time-scale of the measurement). In some examples, the sample can include a R2R web.

As a rough estimate, typical R2R speeds for high resolution patterning are 10 cm/s. For a spot size diameter of ~300 μm, this allows a measurement time of 3×10^{-3} sec before the measurement spot moves to a completely new area.

The present 2θ-scatterometer apparatus allows for optical beam scanning without interference with a sample support surface or movement of the sample by the support surface. The general concept of the present disclosure is: to focus a polarized laser beam with a optical system characterized by a first numerical aperture (NA₁) onto a galvanometer mirror; to rotate the galvanometer mirror across a second $\sin^{-1}(\text{NA}_2)$ angular range, $\sin^{-1}(\text{NA}_2) = \sin^{-1}(\text{NA}_2)_{\max} - \sin^{-1}(\text{NA}_2)_{\min}$ where $\sin^{-1}(\text{NA}_2) > \sin^{-1}(\text{NA}_1)$; to image the galvanometer mirror focal spot with NA₂ optics onto a conjugate image at the sample surface; and to relay the image further onto a detector with NA₂ optics, wherein the detector surface is also a conjugate image of the galvanometer mirror first surface and the surface of the sample. The NA₂ optics can constitute spherical lenses, spherical mirrors or paraboloidal mirrors.

Scatterometry measures the reflectivity from a planar sample which may contain a periodic array of micro- to nano-scale structures. The periodic array can be one-dimensional or two dimensional in the plane of the sample, and can have structure in the direction perpendicular to the surface (for a one-dimensional array, the structure could be a simple line with a width and a height above the surface, e.g. a grating, and the perpendicular structure could be the height of the lines). More complex perpendicular structures such as a layered structure, or a structure for which the line: space ratio changes with height are often encountered in manufacturing. Metrology is required to monitor the variations in these structure characteristics as the manufacturing process proceeds.

For periodicities smaller than $\lambda/2$, there are no propagating diffracted orders other than the zero-order reflected beam when a light is incident on the sample. However, a reflectivity vs. angle measurement provides information on the details of the structure.

There are several parameters of the measurement that can be varied to provide information on the sample. These include: 1) the polarization of the incident light. Most often for 2θ scatterometry linearly polarized light is used, polarized either in the plane of incidence (TM polarization) or perpendicular to the plane of incidence (TE polarization). The plane of incidence is the plane formed by the normal to

the sample and the direction of the incident beam. In certain cases, circular polarization may provide additional information. Thus, it is necessary to build polarization flexibility into the measurement system to allow optimization for different samples.

For a one-dimensional array, the array wavevector (a vector in the plane of the sample perpendicular to the lines of the array) can be at any angle φ relative to the plane of incidence. For most familiar grating applications, $\varphi=0^\circ$. If $\varphi \neq 0^\circ$, the geometry is known as a conical diffraction geometry. Most often in 2θ scatterometry reflectance curves are measured for $\varphi=0^\circ$ and for $\varphi=90^\circ$; but other angles can be investigated and may contain additional information for some samples.

For a two-dimension array, there are two (or more depending on the complexity of the unit cell) array wavevectors. Measurement protocols will be specific for different sample arrays. As always, there is an engineering trade-off between simplicity of design and flexibility of experimental configuration. For most samples, it is sufficient to consider linear polarization (TE or TM) and $\varphi=0^\circ$ or 90° .

FIG. 2 shows a first in-line optical scatterometry system 200 according to examples of the present disclosure. A collimated laser source at a fixed wavelength (λ) 202 produces an optical beam 204, which is directed to a polarization control element 206 that transmits the optical beam 208 with a controlled polarization. In an embodiment, the first optical beam can have a collimated spot size of about 4 mm². A first lens 210 focuses the optical beam 212 onto a surface 214 of a first-surface galvanometer mirror 216. The galvanometer mirror 216 is configured to actuate through a range of angles to reflect the optical beam 218 with a direction that changes with the rotation of the galvanometer mirror over a range of directions that span across an entrance surface 220 of a second lens 222. The second lens 222 approximately collimates and transmits the optical beam 224 that is directed to a first reflecting surface 226 of a first parabolic mirror 228. The first reflecting surface of the paraboloidal mirror 226 reflects the beam 224 to produce an optical beam 230 that is incident on a first surface 234 of a sample 232 at a target location 234. In some examples, by using the combination of the first lens 210, the second lens 222, and the first parabolic mirror 228, the optical beam 230 can have a dimension of about $240 \times 115 \mu\text{m}^2$ at the target location 234 in the center of the angular scan. This spot size will vary across the angular range since the distance from the parabolic mirror to the sample varies with the angle of incidence. For 2θ scatterometry, the reflectance is independent of the spot size as long as the spot is much larger than the array period. The target location, which can be a surface of a R2R web, is optically conjugate to the front surface of the galvanometer mirror. As the galvanometer mirror 216 rotates, the optical beam 224 is swept across the first reflecting surface 226, which images the target location 234 with a time-varying angle of incidence. In an embodiment, the angular range produced by the galvanometer mirror 216 can be between about 20° to about 85° . A front reflecting surface 236 of a second parabolic mirror 238 receives the optical beam reflected from the sample surface 230 at position 234 as optical beam 240. For specular reflection from a planar surface, the angle of incidence of the incident beam relative to a normal to the surface is equal to the angle of reflection (both indicated by θ in the figure). The second parabolic first surface mirror 238 reflects and collimates the optical beam 240 as optical beam 242 to an entrance surface 244 of a third lens 246. The third lens 246 focuses the optical beam 242 as optical beam 248 to a detector 250. The second

parabolic mirror **238** and the third lens **246** are arranged such that the optical beam **248** remains focused at the detector **250** during the scanning operation. The arrangement of FIG. **2** is such that a detection surface of the detector **250** is conjugate to the focal point of the laser source beam onto the galvanometer mirror **216**. The detector **250** can be coupled with a signal processing electronics **254** to process a signal detected by the detector **250** to perform a measurement process to measure the angular dependence of the light reflected from target location **234**. For example, for a sinusoidal galvanometer oscillation at 2.0 kHz, a scan duration of about ~0.25 ms is recorded (at 2.0 kHz the cycle repeats every 0.5 ms, but a cycle includes both a trace and a retrace) with the signal processing electronics **254**, which can provide a reduction in the time for scatterometry signal acquisition as compared with the prior art. The signal processing electronics can receive and process several scans, which can be averaged as necessary for improved signal to noise. The goal is to have the measurement time comparable to the transit time of the web across the spot on the sample surface. Typical web scan speeds are up to 10 cm/s. Thus, the time for the web to move across a ~300 μm diameter spot is ~3 ms, at a scan speed of 2 kHz, the time necessary for a scan is 250 μsec [$\frac{1}{2}$ of the (repetition rate) $^{-1}$] which allows a comfortable margin for averaging multiple scans to improve signal-to-noise. Higher scan speeds are available as needed.

Scatterometry measures the reflection from a periodic structure. Unlike microscopy, this is an ensemble measurement that does not measure specific individual features, but measures the reflectivity from a group of features and uses an inverse modeling approach to deduce the feature dimensions and structure. For a manufacturing application, where the range of feature dimensions likely to be encountered is bounded, a simple approach to the fitting is to pre-calculate the forward problem (scattering intensity vs. angle) for a range of acceptable feature sizes and structure shapes and then evaluate the measured spectrum by a rapid least squares fitting to the pre-calculated angular spectra. This procedure can be configured to provide an alarm if the measured spectra deviate from an acceptable variation. A potential issues with scatterometry system **200** is that the optics extend below the sample surface. For a small test sample, this is not an issue. However, for integration with a R2R manufacturing system, the path of the web might interfere with the optical path as shown.

FIG. **3A** shows a second in-line optical scatterometry system **300** according to examples of the present disclosure. FIG. **3B** shows a perspective view of a portion of the optical set-up of FIG. **3A** showing the mirror arrangement near the parabolic mirror. This system is designed to avoid the optical path issues discussed in conjunction with system **200**. The system **300** is configured to measure a planar device **302**. The system **300** comprises a first optical component **304** that is configured to produce an optical beam **306**. For example, the first optical component **304** comprises the output of a laser **308** with appropriate optical components to result in a collimated beam. The diameter of the collimated beam may be ~2 mm. The system **300** further comprises a second optical component **309** that is configured to receive the optical beam **306** and produce a first conditioned optical beam **310**. For example, the second optical component **309** comprises one or more optical elements configured to polarize and focus the optical beam produced by the first optical component **304** onto a front surface of moving mirror **316** via mirror **311**. For example, the first conditioned optical beam **310** can have a beam profile having dimensions of about ~0.14 mm² at the front surface of the moving mirror

316. The NA₁ at the moving mirror **316** can be about 0.004. The moving mirror **316** is configured to receive the first conditioned optical beam **310** and sweep the first conditioned optical beam **310** across range of angles commensurate with the numerical aperture NA₂ of the following optical components. For example, the movable mirror **316** is configured to move at a frequency of about 2 kHz, which results in a scanning time of about 0.25 msec/half-scan. For example, the moveable mirror **316** can be a galvanometer mirror that includes mechanical and electrical resonance structures designed for a specific frequency. For example, if the sample is a web that is moved during the measurement process, the frequency at which the movable mirror **316** is moved can be varied based on a speed at which the web is moved. For a web movement speed of 10 cm/sec, the overall measurement process can be about 5 msec.

The system **300** further comprises additional optical components that are configured to receive the conditioned optical beam **310** from the movable mirror **316** and direct the conditioned optical beam **310**. For example, the additional optical components comprises collimating lens **320**, a trio of flat mirrors **322**, **324** and **326** and a first parabolic mirror **332** that is configured to receive the conditioned optical beam **310** from the movable mirror **316** and direct the conditioned optical beam **310** to a focus conjugate with the focus at the movable mirror **316**, i.e., galvanometer mirror, at target area **338** of the planar device **302**. The flat mirror **326** allows for the conditioned optical beam **310** to be directed from a side position of the target area **338** and avoids any interference between the optics and the web path. The first parabolic mirror **332** can be a 45° off-axis parabolic mirror that allows for a scanning range of about 50° for the conditioned optical beam **310** at the target area **338** (e.g. this is an off-axis optical system with θ ranging from 15° to 65° or NA_{2min}=0.26 and NA_{2max}=0.906). This range can be adjusted by variation of the optical system, for example the range can be adjusted to 30° to 80° (NA_{2min}=0.5 and NA_{2max}=0.985). A higher scanning range is available with different parabolic mirrors. The system **300** further comprises a collection optical system that is configured to receive a reflected optical beam **340** from the target area **338**. For example, the collection optical system comprises a second parabolic mirror **344**, flat mirrors **354**, **355** and **356**, and focusing lens **358**.

For example, the movable mirror **316** is configured to sweep the first conditioned optical beam **310** across a surface of the first parabolic mirror **332** to scan the focal spot across the semiconductor device **302** with a time-varying angle of incidence. The first conditioned optical beam **310** can be swept across the surface of the first parabolic mirror **332** over an angular range of about 30° to about 65°.

The system **300** further comprises a detector **362** that is configured to receive the conditioned scattered optical beam **360** that is used to measure a feature of the target area **338**. As in the previous system of FIG. **2**, the front surface of the movable mirror **316**, the sample area **338** and the detector **362** are optically conjugate surfaces that are invariant as the angle is scanned. The total volume occupied by the various optical components of the system **300** can be about ~0.12 m³. The detector **362** can be in data communication with an oscilloscope **364** and signal processor of a computer **366** to process signals from the detector **362**. For a sinusoidal galvanometer oscillation of the movable mirror **316** of 2 kHz, a scan duration of about 0.25 ms is recorded by the detector **362** and processed by the oscilloscope **364** and computer **366**. The oscilloscope **362** can average several scans for improved signal-to-noise ratio.

FIG. 4 shows a three dimensional (3D) view of system 400 that is similar to the system of 300. FIG. 4 also show relevant dimensions between various optical components of system 400. In FIG. 4, two scatterometry systems operating in orthogonal directions are shown to provide additional metrology information. In FIG. 4, two lasers are shown that can operate simultaneously or in sequence to perform different measurements of a sample surface. Optical beams produced by a first laser 402 and a second laser 404 are directed to a first mirror 406 and a second mirror 408, respectively. Optical beams reflected by the first mirror 406 and the second mirror 408 are directed to a first movable mirror 410 and a second moveable mirror 412, respectively. In the case of the optical beam reflected by the second mirror 408, a third mirror 411 is also used. Optical beams reflected by the first movable mirror 410 and the second movable mirror 412 are directed to a set of three (3) flat mirrors 414 and a set of three flat mirror 416, respectively. Optical beams reflected the flat mirrors 414 and the flat mirrors 416 are directed to first parabolic mirrors 418 and second parabolic mirrors 420. Scattered optical beams from the sample surface 422 are reflected by a third parabolic mirror 424 and a fourth parabolic mirror 426 and directed to a set of three flat mirror 428 and a second set of three flat mirror 430, respectively. Optical beams reflected from flat mirrors 428 and the flat mirrors 430 are directed to detector 432 and detector 434. The distance between the first laser 402 and the second laser 404 and the sample surface 422 (height of the system 400) can be about ~600 mm. The width of the system 400 can be about ~320 mm. The depth of the system 400 can be about ~610 mm. The entire system can be translated perpendicularly to the web direction, as indicated by arrow 436, to provide information on structures at various positions across the web with a total system 400 volume of ~0.12 m³.

System 400 can be configured to move side to side in the direction perpendicular to the web motion to provide information on the performance of the roll-to-roll system across the web.

FIG. 5 show a method 500 for measuring an angular spectrum from a periodic array of structures and uses an inverse fitting process to deduce the feature dimensions. The optical system is configured as shown in FIGS. 2 and 3A and 3B. The method 500 begins by collimating and polarizing, at 502, using a first optical system, an optical beam from a laser light source to produce a collimated and polarized optical beam. For example, with regard to FIG. 2, the collimated laser source 202 produces the first optical beam 204 that is polarized by the polarized element 206 and focused by the first lens 210. For example, with regard to FIG. 3, the optical beam 306 produced by the laser 308 is conditioned by the second optical component 309 by polarizing and focusing the optical beam 306.

The method 500 continues by projecting, at 504, using a second optical system, a focal spot of the collimated and polarized optical beam onto a sample surface with a dynamically variable angle of incidence. For example, with regard to FIG. 2, the galvanometer mirror 216, the second lens 222, and the first parabolic mirror 228 are used to direct the optical beam 230 to the target location 234. For example, with regard to FIG. 3, the conditioned optical beam 310 is swept through a range of angles by the movable mirror 316, which is then received by the additional optical components, e.g., collimating lens 320, flat mirrors 322, 324, 326, and the first parabolic mirror 332 332 to be directed onto the planar sample 302 at a target area 338.

The method 500 continues by collecting, at 506, by additional optical components, light reflected from the sample at an optical detector. For example, with regard to FIG. 2, the second parabolic mirror 238 and the third lens 246 receive a reflected optical beam from the target location 234 and direct the eighth optical beam to the detector 250. For example, with regard to FIG. 3, the flat mirrors 354, 355, 356 and the focusing lens 358 collect and focus the scattered optical beam 360 from the target area 338.

The method 500 continues by recording, at 508, a signal from the optical detector. For example, with regard to FIG. 2, the detector 250 receives the optical beam 248 and provides one or more signals indicative of the intensity of optical beam 248 as a function of the polarization and the angle of incidence to the computer 252 and/or the digital oscilloscope 254 for processing to measure a characteristic features of the sample 232 at the target location 234. For example, with regard to FIG. 3, the detector 362 receives optical beam 360 from the second parabolic mirror 344, the flat mirrors 354, 355, 356, and the focusing lens 358. The detector 362 can be in data communication with the oscilloscope 364 and/or the computer 366. The computer 366 averages multiple scans and performs a data fitting operation to measure a feature of the target area 338.

For example, the computer 252 or computer 366 can be configured to perform operations described in Zhu, A. Munoz, S. R. J. Brueck, S. Singhal, S. V. Sreenivasan, *Metrology of a 50-nm HP wire-grid polarizer: a SEM-scatterometry comparison*, Proc. SPIE 9424, 9424IW (2015) and R. Zhu, S. R. J. Brueck, N. Dawson, T. Busani, P. Joseph, S. Singhal and S. V. Sreenivasan, *Scatterometry for Nanoimprint Lithography*, J. Vac. Sci. Tech. B34, 06K503 (2016). Several simulation techniques can be used for the forward analysis (computing the reflectivity vs. angle for the given polarization and conical diffraction conditions for a fixed structure). One embodiment uses rigorous coupled wave analysis (RCWA); finite difference time domain (FDTD) approaches are another embodiment, finite element analysis (FE) is a third embodiment.

In one example, a rigorous coupled-wave analysis with a simulation model of a periodic array of Al lines on top of a fused silica substrate is used by the computer 252 or the computer 366. From SEM results the etching of the Al lines extended slightly into the silica substrate so an additional parameter, the depth of a SiO₂, etch is added to the model. The grating profile is defined by seven parameters: pitch (P), bottom linewidth (LW), top linewidth (TW), Al thickness (Al), fused silica undercut (FS), horizontal (HR) and vertical extent (VR) of the top rounding. The refractive indices of Al and SiO₂ are taken from the literature. For the analysis, each term of the seven parameter is individually varied, while other terms are held constant. A starting point for the analysis is chosen to be a reasonable starting guess for each of the seven parameters by considering a cross section of a SEM image of the sample. Using this analysis of the simulation results, wavelengths including 244 nm, 405 nm and 633 nm have enough capability and sensitivity to resolve 100-nm periodic structures.

FIG. 6 shows an example for a wire grid polarizer structure (a nominally 100-nm pitch one-dimensional Al grating structure on a fused silica substrate. FIG. 6 is taken from FIG. 11 of Zhu et al. The manufacturing process includes depositing a blanket Al film of nominally 200 nm thickness, nanoimprint lithography to define the grating structure, reactive-ion etching to remove the Al between the grating lines, which results is some rounding of the top of the lines as well as a taper of the linewidth from top to bottom.

As a result of the etching process there is a small undercut into the fused silica substrate. Additionally an Al oxide overcoat on the Al forms on exposure to air. Thus, there are eight parameters that define even this straightforward manufacturing process: 1) the period of the grating, P; 2) the bottom Al linewidth (at the fused silica interface), BW; 3) the top Al linewidth (before the rounding), TW; 4) the thickness of the Al line, H; 5) the horizontal rounding (HR); 6) the vertical rounding, VR; 7) the undercut into the fused silica, SIO; and 8) the ALO thickness, ALO. Two experimental geometries were measured with the apparatus of FIG. 1. A) TE polarization with the grating lines perpendicular to the axis of polarization, and B) TM polarization with the grating lines parallel to the axis of rotation. As can be seen in FIG. 6, there is excellent agreement between the measured and simulated angular reflectivities for the parameters: P=97.2 nm; BW=49.4 nm; TW=58.6 nm; H=208.3 nm; HR=25.8 nm; VR=14.8 nm; SIO=22.9 nm; and ALO=4 nm.

The computer 252 or the computer 366 can be implemented with a general purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor can be a microprocessor, but, in the alternative, the processor can be any conventional processor, controller, microcontroller, or state machine. A processor can also be implemented as a combination of computing devices, e.g., a combination of a DSP and/or GPU and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core and/or GPU core, or any other such configuration.

In one or more exemplary embodiments, the functions described can be implemented in hardware, software, firmware, or any combination thereof. For a software implementation, the techniques described herein can be implemented with modules (e.g., procedures, functions, subprograms, programs, routines, subroutines, modules, software packages, classes, and so on) that perform the functions described herein. A module can be coupled to another module or a hardware circuit by passing and/or receiving information, data, arguments, parameters, or memory contents. Information, arguments, parameters, data, or the like can be passed, forwarded, or transmitted using any suitable means including memory sharing, message passing, token passing, network transmission, and the like. The software codes can be stored in memory units and executed by processors. The memory unit can be implemented within the processor or external to the processor, in which case it can be communicatively coupled to the processor via various means as is known in the art.

In some examples of the present disclosure defects of the sample structure can be analyzed as discussed below. Defects are typically observed by a change in the reflectivity of the sample as a function of the position on the sample and the angle of incidence. In one example, wherein the beam is focused onto the sample, the defect is detected as a change in the reflectivity vs angle of incidence. If the beam scan is sufficiently fast that the scan is completed before the sample in the R2R has moved by a beam spot size, a single region of the sample at the focal spot is observed. Alternatively, if the beam scan is adjusted, for example, by moving the sample out of the focal position of the optical system, both the angle of incidence and the position on the sample are varying during the scan.

For R2R manufacturing processes, the incident optical system, such as the systems discussed above, can be adjusted to be slightly out of focus which will allow sweeping of the beam by parabolic mirror 728 (or parabolic mirror 228 of FIG. 2) to scan across the sample 732, such as a web, so that both the position of the beam and the angle of incidence are simultaneously varied at the sample 732 or web during the scanning. The reflectivity will show a discontinuous behavior as the beam scans across the defect as a result of the differing optical properties of the defect as compared to the non-defected structure/material. In other cases, with different structures and defect modes, the reflectivity may decrease rather than increase. The important feature is the rapid change in the reflectivity as the beam sweeps across the defect. These signal variations can be associated with defects on the web. This will allow the system to monitor multiple aspects of the structure on the web: critical dimensions and defects. For the optical system as shown in FIG. 2, the angular scan is from $\sim 29^\circ$ to $\sim 59^\circ$ and the position scan across the sample, set by the defocus distance is ~ 500 micrometers. These dimensions can be changed with different optical systems and defocus distances as is well known in the art. FIG. 7A presents a focused configuration in which the spot on the web is more or less fixed as the angle of incidence is swept. FIG. 7B shows an alternate configuration where the web has been moved closer to the web so that both the angle of incidence and the position on the web are swept in the same scan. Alternatively, the web can be moved down so that the focus is above the web. Often the web speed is slower than the scan speed, so the web can be considered almost stationary during a sweep. If a defect is in the sweeping area, the reflectivity measurements will show a rapid variation as the beam scans across the defect. Since the reflectivity is a relatively slow function of the incidence angle, dimensional and defect measurements can be combined into a single trace and separated by electronic filtering.

FIG. 8A and FIG. 8B show observation of a defect during an in-line optical scatterometry measurement as the optical beam is swept across a $50\text{ }\mu\text{m}$ wide non-filling defect according to examples of the present disclosure. As shown in FIG. 7B, the beam is scanned across the R2R web by adjusting the focal length of the parabolic mirrors or the relative positions of the parabolic mirror and the web to be slightly out-of-focus according to examples of the present disclosure. The beam focus is adjusted so that the beam sweeps across a region of the sample so that both the position and the angle of incidence are scanned simultaneously. This sweeping or scanning process will allow rapid scanning of defects by collecting the defect reflection signature variations. The reflectivity variations will allow the system to characterize defects size and types (non-filling, pinholes, striations, etc.). FIG. 8A presents a relatively large non-filling defect ($\sim 50\text{ }\mu\text{m}$; the kerf region). FIG. 8B shows the spike observed on the oscilloscope trace that represents the signal detection from the beam sweeping across the defect. An advantage of the position sweeping is that the defect shows up as a rapid change in the reflectivity which can be isolated from the structure information and detected with simple electronic filtering. The sensitivity to defects will depend on both the change in the reflectivity parameters for the defect as compared with the filled region, and the size of the defect. Simply from the example shown in FIGS. 8A and 8B, the signal to noise (S/N) is about 10 as the defect is scanned. This suggests that defects of ~ 5 micrometer in the

direction of the web motion will be readily detected. With additional signal processing, this scale can be reduced to sub-micrometer scales.

FIG. 9 shows a method 900 for high-speed 2θ scatterometry defect detection according to examples of the present disclosure. The method 900 can use the optical system as configured and shown in FIG. 2, FIG. 3A, FIG. 3B, and FIG. 7A and FIG. 7B. The method 900 comprises collimating, using one or more collimating optical elements, a beam from a laser light to produce a collimated beam, as in 902. The method 900 continues by polarizing, using one or more polarizing optical element, the collimated beam to produce a polarized and collimated beam, as in 904. For example, with regard to FIG. 2, the collimated laser source 202 produces the first optical beam 204 that is polarized by the polarized element 206 and focused by the first lens 210. For example, with regard to FIG. 3, the optical beam 306 produced by the laser 308 is conditioned by the second optical component 309 by polarizing and focusing the optical beam 306.

The method 900 continues by directing the polarized and collimated beam, by a first optical system, onto a surface of a sample with a dynamically variable angle of incidence and dynamically variable position, as in 906. For example, with regard to FIG. 2, the galvanometer mirror 216, the second lens 222, and the first parabolic mirror 228 (or parabolic mirror 728) are used to direct the optical beam 230 to the target location 234. For example, with regard to FIG. 3, the conditioned optical beam 310 is swept through a range of angles by the movable mirror 316, which is then received by the additional optical components, e.g., collimating lens 320, flat mirrors 322, 324, 326, and the first parabolic mirror 332 to be directed onto the planar sample 302 at a target area 338. Also, with regard to FIG. 7A and/or FIG. 7B, the parabolic mirror 728, such as the parabolic mirror 228 of FIG. 2, is configured to direct the polarized and collimated beam to the surface of a sample with a dynamically variable angle of incidence and dynamically variable position.

The method 900 continues by collecting, by a second optical system, reflected light from the sample, as in 908. The method 900 continues by projecting the reflected light onto an optical detector, as in 910. The method 900 continues by recording the reflected light that is collected by the second optical system, as in 912. The method 900 continues by analyzing the reflected light that is recorded by a signal analyzer for signal analysis and averaging and that provides a scatterometry measurement of local dimensions of a structure of the sample and a detection of defects on the sample, as in 914. For example, with regard to FIG. 2, the second parabolic mirror 238 and the third lens 246 receive a reflected optical beam from the target location 234 and direct the reflected optical beam to the detector 250. For example, with regard to FIG. 3, the flat mirrors 354, 355, 356 and the focusing lens 358 collect and focus the scattered optical beam 360 from the target area 338. Also, with regard to FIG. 2, the detector 250 receives the optical beam 248 and provides one or more signals indicative of the intensity of optical beam 248 as a function of the polarization and the angle of incidence to the computer 252 and/or the digital oscilloscope 254 for processing to measure characteristic features of the sample 232 at the target location 234. For example, with regard to FIG. 3, the detector 362 receives optical beam 360 from the second parabolic mirror 344, the flat mirrors 354, 355, 356, and the focusing lens 358. The detector 362 can be in data communication with the oscilloscope 364 and/or the computer 366. The computer 366

averages multiple scans and performs a data fitting operation to measure a feature of the target area 338.

Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the embodiments are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in their respective testing measurements. Moreover, all ranges disclosed herein are to be understood to encompass any and all sub-ranges subsumed therein. For example, a range of “less than 10” can include any and all sub-ranges between (and including) the minimum value of zero and the maximum value of 10, that is, any and all sub-ranges having a minimum value of equal to or greater than zero and a maximum value of equal to or less than 10, e.g., 1 to 5. In certain cases, the numerical values as stated for the parameter can take on negative values. In this case, the example value of range stated as “less than 10” can assume negative values, e.g. -1, -2, -3, -10, -20, -30, etc.

The following embodiments are described for illustrative purposes only with reference to the Figures. Those of skill in the art will appreciate that the following description is exemplary in nature, and that various modifications to the parameters set forth herein could be made without departing from the scope of the present embodiments. It is intended that the specification and examples be considered as examples only. The various embodiments are not necessarily mutually exclusive, as some embodiments can be combined with one or more other embodiments to form new embodiments.

While the embodiments have been illustrated respect to one or more implementations, alterations and/or modifications can be made to the illustrated examples without departing from the spirit and scope of the appended claims. In addition, while a particular feature of the embodiments may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular function.

Furthermore, to the extent that the terms “including”, “includes”, “having”, “has”, “with”, or variants thereof are used in either the detailed description and the claims, such terms are intended to be inclusive in a manner similar to the term “comprising.” As used herein, the phrase “one or more of”, for example, A, B, and C means any of the following: either A, B, or C alone; or combinations of two, such as A and B, B and C, and A and C; or combinations of A, B and C.

Other embodiments will be apparent to those skilled in the art from consideration of the specification and practice of the descriptions disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit of the embodiments being indicated by the following claims.

What is claimed is:

1. A method for 2θ scatterometry defect detection comprising:
 - providing a beam from a laser light source;
 - providing collimation and polarization control optics for the beam from the laser light source;
 - providing a first optical system configured to project the beam from the laser light source onto a surface of a sample with a dynamically continuous variable angle of incidence and dynamically variable position;

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defocusing the beam at the surface of the sample by the first optical system, wherein the beam that is defocused provides a laser spot on the sample that scans across the sample with the dynamically continuous variable angle of incidence;

providing a second optical system configured to collect reflected light from the sample and project the reflected light onto an optical detector;

recording the reflected light that is collected by the second optical system; and

analyzing the reflected light that is recorded by a signal analyzer to provide for signal analysis and signal averaging and that provides a scatterometry measurement of local dimensions of a structure of the sample.

2. The method of claim 1, further comprising projecting a focus of the beam from the laser light source onto the sample by the first optical system.

3. The method of claim 1, wherein the signal analyzer comprises frequency selective circuitry to separately record first frequency variations of a reflectivity signal that correspond to a scatterometry signature sensitive to dimensions of the structure on the surface and second frequency variations of the reflectivity signal that correspond to defects in the structure on the surface, wherein a spectral range of the first frequency variations is lower than a spectral range of the second frequency variations.

4. The method of claim 1, wherein an angular scan performed by the first optical system is from approximately 29° to approximately 59°.

5. The method of claim 1, wherein an angular scan performed by the first optical system is from approximately 29° to approximately 59° and a position scan is across a distance of approximately 500 micrometers.

6. A method for 2θ scatterometry defect detection comprising:

collimating, using one or more collimating optical elements, a beam from a laser light source to produce a collimated beam;

polarizing, using one or more polarizing optical element, the collimated beam to produce a polarized and collimated beam;

directing the polarized and collimated beam, by a first optical system, onto a surface of a sample with a dynamically continuous variable angle of incidence and dynamically variable position;

defocusing the beam at the surface of the sample by the first optical system, wherein the beam that is defocused provides a laser spot on the sample that scans across the with the dynamically continuous variable angle of incidence;

collecting, by a second optical system, reflected light from the sample;

projecting the reflected light onto an optical detector;

recording the reflected light that is collected by the second optical system; and

analyzing the reflected light that is recorded by a signal analyzer to provide for signal analysis and signal averaging and that provides a scatterometry measurement of local dimensions of a structure of the sample.

7. The method of claim 6, further comprising projecting a focus of the beam from the laser light source onto the sample by the first optical system.

8. The method of claim 6, wherein the signal analyzer comprises frequency selective circuitry to separately record

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first frequency variations of a reflectivity signal that correspond to a scatterometry signature sensitive to dimensions of the structure on the surface and second frequency variations of the reflectivity signal that correspond to defects in the structure of the surface, wherein a spectral range of the first frequency variations is lower than a spectral range of the second frequency variations.

9. The method of claim 6, wherein an angular scan performed by the first optical system is from approximately 29° to approximately 59°.

10. The method of claim 6, wherein an angular scan performed by the first optical system is from approximately 29° to approximately 59° and a position scan is across a distance of approximately 500 mm.

11. A system for 2θ scatterometry defect detection comprising:

a laser light source that produces a beam;

one or more control optics that produce a collimated and polarized beam;

a first optical system that projects the collimated and polarized beam from the laser light source onto a surface of a sample with a dynamically continuous variable angle of incidence and dynamically variable position, wherein the one or more control optics defocuses the collimated and polarized beam at the surface of the sample by the first optical system, wherein the beam that is defocused provides a laser spot on the sample that scans across the sample with the dynamically continuous variable angle of incidence;

a second optical system that collects reflected light from the sample and projects the reflected light onto an optical detector;

a signal recorder that records the reflected light that is collected by the second optical system; and

a signal analyzer that analyzes the reflected light that is recorded to provide for signal analysis and signal averaging and that provides a scatterometry measurement of local dimensions of a structure of the sample.

12. The system of claim 11, wherein the one or more control optics projects a focus of the collimated and polarized beam from the laser light source onto the sample by the first optical system.

13. The system of claim 11, wherein the signal analyzer comprises frequency selective circuitry to separately record first frequency variations of a reflectivity signal that correspond to a scatterometry signature and dimensions of the structure of the surface and second frequency variations of the reflectivity signal that correspond to defects in the structure of the surface, wherein a spectral range of the first frequency variations is lower than a spectral range of the second frequency variations.

14. The system of claim 11, wherein an angular scan performed by the first optical system is from approximately 29° to approximately 59°.

15. The system of claim 11, wherein an angular scan performed by the first optical system is from approximately 29° to approximately 59° and a position scan is across a distance of approximately 500 mm.

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